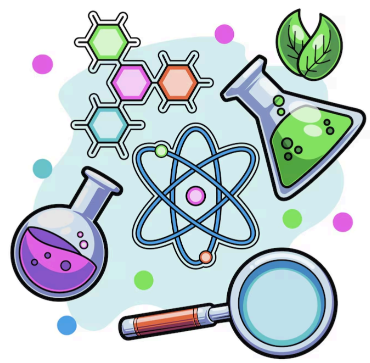
UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

District • 2026



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

B01. In mice, black fur (B) is dominant to brown fur (b), and short tails (T) are dominant to long tails (t). A mouse that is heterozygous for both traits is crossed with a mouse that is homozygous for black fur and has a long tail. What percentage of the offspring from the following cross would phenotypically express black fur and short tails?
A) 0%
B) 25%
C) 50%
D) 75%
E) 100%

B02. A dehydration (condensation) reaction would not occur during the formation of a(n)
A) Disulfide bond.
B) Peptide bond.
C) Glycosidic bond.
D) Ester linkage.
E) Disaccharide from two monosaccharides.

B03. Two organisms share the same genus but not the same species. What can most reasonably be inferred?
A) They belong to different domains.
B) They cannot have any evolutionary relationship.
C) They must live in the same ecosystem.
D) They are classified as the same organism.
E) They likely share many structural and genetic similarities.

B04. If a cell bypasses the M checkpoint and enters anaphase with some chromosomes not attached to spindle fibers, which of the following consequences is most likely?
A) Premature entry into S phase.
B) Incomplete formation of the cleavage furrow.
C) Immediate withdrawal from the cell cycle.
D) Unequal distribution of genetic material to daughter cells.
E) Increased rate of apoptosis in metaphase.

B05. According to the World Health Organization (WHO), on January 30, 2026, the Ministry of Health of Ethiopia declared the end of an outbreak of which viral disease?
A) Marburg virus disease (MVD)
B) Seasonal influenza (A(H3N2))
C) Middle East respiratory syndrome (MERS)
D) Ebola virus disease (EVD)
E) Chikungunya virus disease (CHIKV)

B06. The complete genetic makeup of an organism is referred to as its
A) chromosome.
B) genome.
C) allele.
D) locus.
E) phenotype.

B07. In which environment would asexual reproduction most likely provide a greater reproductive advantage than sexual reproduction?
A) An environment with rapidly changing pathogens.
B) A stable environment with minimal long-term environmental change.
C) An environment experiencing frequent population bottlenecks.
D) An environment with strong selective pressure favoring rare alleles.
E) An environment with high parasite diversity.

B08. Which feature is most characteristic of the paroxysmal stage of pertussis?
A) Low-grade fever and malaise.
B) Progressive shortness of breath.
C) Productive cough with green sputum.
D) Rash and swollen lymph nodes.
E) Severe, repetitive coughing fits followed by a "whoop" sound.

B09. Which statement best describes the role of water in the light-dependent reactions of photosynthesis?
- A) Water is oxidized to store energy in oxygen molecules.
- B) Water donates electrons to photosystem II.
- C) Water provides protons for carbon fixation.
- D) Water stores energy in chemical bonds.
- E) Water directly drives ATP synthase.

B10. The type of epithelial tissue that forms a single layer of flattened cells is
- A) simple columnar.
- B) simple cuboidal.
- C) stratified columnar.
- D) simple squamous.
- E) stratified squamous.

B11. The domain that includes single-celled organisms that are often found in very extreme environments (high salt, high pressure, low pH, very hot or cold temperatures), and are not known to be pathogenic is Domain
- A) Bacteria.
- B) Archaea.
- C) Eukarya.
- D) Protista.
- E) Prokarya.

B12. How do allosteric regulators affect enzyme activity?
- A) By competing with the substrate for occupancy of the active site.
- B) By causing permanent denaturation under physiological conditions.
- C) By binding at a site other than the active site and altering the enzyme’s catalytic function.
- D) By increasing substrate availability to drive the reaction forward.
- E) By changing the overall free energy change (ΔG) of the reaction.

B13. Which organ systems are most directly involved in regulating blood glucose levels shortly after a meal?
- A) Digestive and endocrine
- B) Nervous and muscular
- C) Endocrine and urinary
- D) Muscular and digestive
- E) Skeletal and immune

B14. What is the role of the operator in an operon?
- A) Serves as a regulatory DNA sequence that controls transcription of structural genes.
- B) Encodes a protein that inhibits RNA polymerase activity.
- C) Initiates transcription by binding ribosomes.
- D) Splices introns out of pre-mRNA.
- E) Directs transport of RNA molecules to the cytoplasm.

B15. In a freshwater ecosystem, zooplankton feed on phytoplankton, and small fish feed on zooplankton. Larger predatory fish feed on the small fish. The small fish are best classified as which of the following?
- A) Producer
- B) Primary consumer
- C) Secondary consumer
- D) Tertiary consumer
- E) Apex predator

B16. A polypeptide has the following amino acid sequence: Met-Ala-Gln-Arg-Glu-Leu. The DNA encoding the polypeptide was mutated, which produced the following mutant sequence: Met-Ala. If sequencing shows the codons downstream of the mutation are unchanged, which describes the most likely type of mutation that occurred?
- A) Neutral mutation
- B) Missense mutation
- C) Frameshift mutation
- D) Reverse mutation
- E) Nonsense mutation

B17. A patient has a disorder caused by a single defective gene. Doctors use a modified harmless virus to deliver a normal version of the gene to the patient's cells to restore normal function. This medical approach is known as:
A) genetic screening.
B) gene therapy.
C) CRISPR DNA sequencing.
D) DNA fingerprinting.
E) stem cell research.

B18. A cell produces large quantities of secreted enzymes. Which organelle is primarily responsible for synthesizing and preparing these proteins?
A) Smooth endoplasmic reticulum
B) Golgi apparatus
C) Lysosome
D) Rough endoplasmic reticulum
E) Plasma membrane

B19. A table shows survival rates of three phenotypes of a butterfly population after a change in environmental conditions:

Phenotype	**Number Before Selection**	**Number After Selection**	**Survival Rate (%)**
Light-colored	100	90	90%
Medium-colored	100	50	50%
Dark-colored	100	10	10%

Which form of selection is this data best illustrating?
A) Disruptive selection
B) Stabilizing selection
C) Sexual selection
D) Directional selection
E) Genetic drift

B20. Predation is similar to ______ in that both types of relationships benefit one of the interacting species while the other is harmed.
A) parasitism
B) commensalism
C) competition
D) mutualism
E) amensalism

C01. How many atoms are there in one formula unit of zinc phosphate?

A) 6 B) 8 C) 13 D) 15 E) 18

C02. What is the insoluble product that is formed when a solution of silver nitrate is mixed with a solution of sodium sulfide?

A) $NaNO_3$ B) AgS C) Ag_2SO_4
D) Ag_2S E) $AgNO_3$

C03. The effective nuclear charge felt by a valence electron in an atom is equal to

A) the atomic number of the element
B) the number of protons and neutrons in the nucleus
C) the number of protons in the nucleus minus the number of inner shell electrons
D) the number of protons in the nucleus minus the number of valence electrons
E) the sum of the protons and neutrons, minus the total number of electrons in the atom

C04. Based on the Lewis dot structure, how many σ bonds and how many π bonds are there in a sulfate ion?

A) 4 σ bonds and 0 π bonds
B) 4 σ bonds and 2 π bonds
C) 4 σ bonds and 4 π bonds
D) 2 σ bonds and 4 π bonds
E) 0 σ bonds and 4 π bonds

C05. When 50.0 mL of water and 50.0 mL of ethanol are mixed at 25°C, the volume of the resulting mixture is 96.4 mL. What is the density of the mixture?

A) 0.926 g/mL
B) 0.895 g/mL
C) 0.955 g/mL
D) 0.872 g/mL
E) 0.818 g/mL

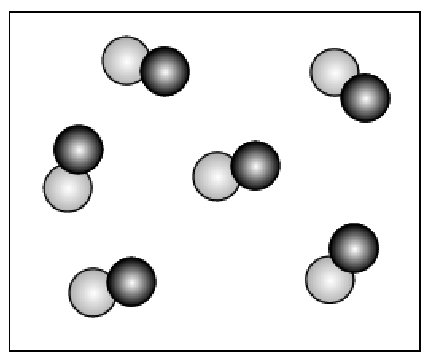
C06. You order a giant 85-liter helium balloon from a sketchy discount party store, but when you pick it up the cashier warns you to keep it away from open flames because they ran out of helium halfway through filling the balloon and they filled it the rest of the way with hydrogen. If the balloon is exactly 85 liters at 1 atm and 23°C, what is the mass of the gas in the balloon?

A) 3.50 g B) 8.76 g C) 10.5 g
D) 17.5 g E) 22.4 g

C07. How much energy is absorbed from the surroundings when 6.5 g of sodium bicarbonate is reacted with excess HCl? ΔH for this reaction is 28.5 kJ/mol.

A) 4385 J B) 2200 J C) 2281 J
D) 2200 kJ E) 2281 kJ

C08. Which term below best describes the gas sample shown here? Each sphere represents one atom.

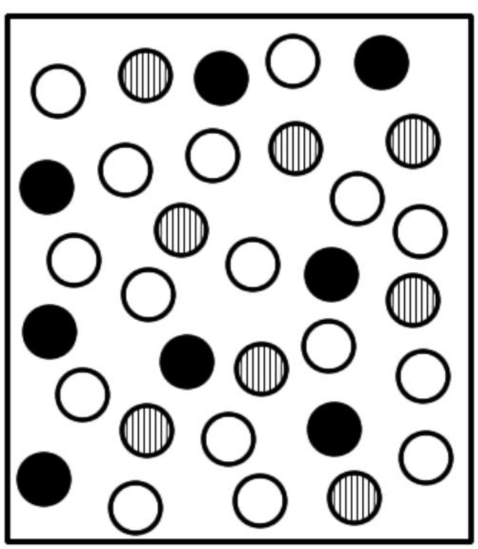


A) Element
B) Compound
C) Mixture of Atoms
D) Mixture of Compounds
E) Mixture of Elements

C09. Imagine you have a block of ice at 0°C and it melts to liquid water at 0°C. Which of the following statements about the phase change is true?

A) The molecules move farther apart when the ice melts to water.
B) The molecules in the liquid water move faster than the molecules in the ice.
C) The ice is colder than the water.
D) The water molecules change from cloudy white to transparent.
E) Heat must have been put into the ice for this to happen.

C10. Below is an image of a gas phase reaction at equilibrium. The equation for the reaction is



If each dot represents 1 mole of gas and the volume of the container is 1000 L, what is K_c? (There are a total of 32 moles of gas in the container at equilibrium.)

A) ½ B) ¼ C) 2 D) 4 E) 16

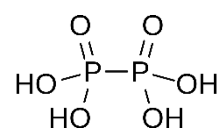
C11. If the equilibrium pH of a 0.50 M solution of a weak acid in 100.0 mL of water is equal to 3.55, what is the K_a for the weak acid?

A) 0.55 B) 0.055 C) 2.26
D) 0.44 E) 1.59×10^{-7}

C12. What is the Ag^+ concentration in a saturated solution of Ag_3PO_4? $K_{sp} = 8.9 \times 10^{-17}$

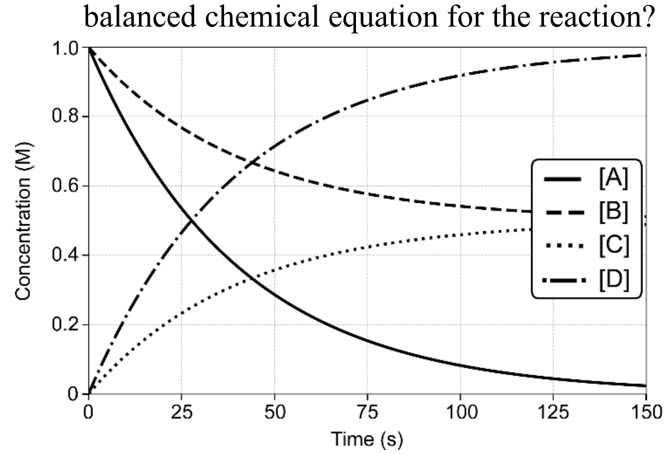
A) 4.26×10^{-5} M
B) 1.27×10^{-4} M
C) 9.72×10^{-5} M
D) 2.54×10^{-4} M
E) 1.42×10^{-5} M

C13. What is the oxidation state on the phosphorus atoms in hypophosphoric acid?



A) 0 B) +2 C) +3 D) +4 E) +6

C14. The following graph shows the change in concentration with time for the species in a certain chemical reaction. What is the balanced chemical equation for the reaction?

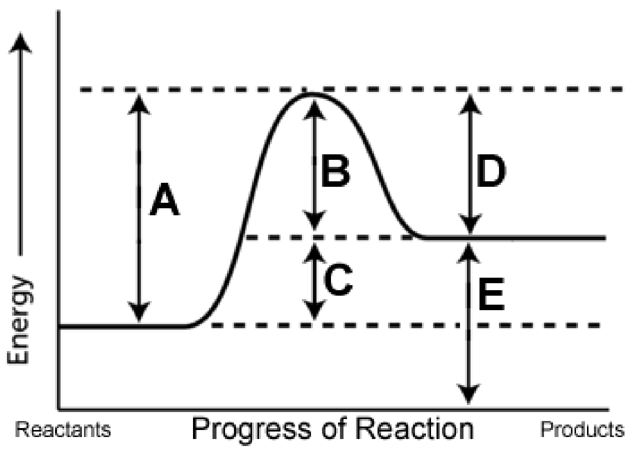


A) A + B → C + D
B) A + D → B + C
C) B + C → 2A + 2D
D) A + 2B → C + 2D
E) 2A + B → C + 2D

C15. A student carrying out an acid-base reaction mistakenly adds 70 mL of 0.010 M HNO_3 to 100 mL of 0.10 M HCl. What is the pH of the resulting solution?

A) 1.20 B) 1.15 C) 1.09
D) 1.27 E) 1.34

C16. In the potential energy diagram shown below, which arrow or arrows tell you whether the forward reaction or the backward reaction has a higher rate constant?



A) A
B) B
C) C
D) A and D
E) A and E

C17. Solid sodium metal reacts with chlorine gas in an exothermic reaction to form solid sodium chloride. What type of reaction is this?

A) Synthesis
B) Decomposition
C) Combustion
D) Precipitation
E) Single Replacement

C18. Which of these compounds has the highest percent iron by mass?

A) Iron(II) nitrate
B) Iron(II) nitrite
C) Iron(II) chloride
D) Iron(II) oxide
E) Iron (II) sulfate

C19. If 1050 mL of 0.47 M sodium bromate is mixed with excess barium nitrate, how many grams of solid barium bromate will be formed?

A) 194 g B) 57 g C) 65 g
D) 73 g E) 97 g

C20. If an electrical current of 0.27 amps is passed through 1.0 L of a 0.50 M solution of $AgNO_3$ for exactly 500 seconds, how many milligrams of solid silver will be formed?

A) 75
B) 120
C) 150
D) 180
E) 210

P01. According to Malley, the observation of "Schweidler's fluctuations" in radioactive decay confirmed that radioactivity is …
A) an electronic, non-nuclear process
B) a nuclear, non-electronic process
C) a clockwork, predictable process
D) a random, probabilistic process
E) a weak force process

P02. According to Malley, in 1911, the idea that each chemical element has a unique atomic weight was discarded. This occurred when a physical chemist proposed that the inseparable elements of mesothorium and radium were not only similar, but identical. Who was this physical chemist?
A) Marie Curie
B) Otto Hahn
C) Frederick Soddy
D) Ernest Rutherford
E) William Marckwald

P03. According to Malley, the discovery that ordinary elements could have different isotopes came from Thompson's lab, when an attempt to separate inert gases produced unexpected curved paths of ions. Thompson's assistant, Francis Aston, determined that the curves were produced by different isotopes of which element?
A) Helium
B) Neon
C) Argon
D) Xenon
E) Radon

P04. A binary star system is observed in which the star light is Doppler shifted as the stars orbit one another, alternating between redshift and blueshift. This kind of binary system is called a(n) ________________.
A) visual binary
B) eclipsing binary
C) spectroscopic binary
D) Kepler binary
E) apparent binary

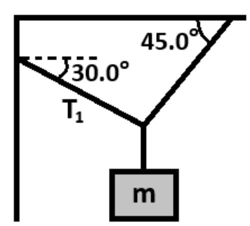
P05. In the formula given below, F is force in Newtons, P is power in Watts, and x is distance in meters. What are the units of the quantity z?

$$x = \frac{Pz}{F}$$

A) seconds
B) meters
C) meters/second
D) kilograms
E) kilograms/second

P06. A football was thrown by a quarterback from a height of 2.0m above the ground. It was caught by a receiver at the same height, 2.0m above the ground. The football was thrown at a speed of 17.0m/s and at an angle of 22.0° above the horizontal. How far down the field from the quarterback was the receiver when he caught the ball?
A) 10.2 m
B) 20.5 m
C) 25.4 m
D) 40.1 m
E) 50.7 m

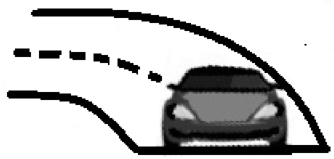
P07. As illustrated below, a 40.0kg mass dangles from a vertical rope that is tied to a pair of cables that connect to the wall and ceiling. The system is in static equilibrium. What is the tension, T_1, in the cable that connects to the wall?
A) 287 N
B) 392 N
C) 432 N
D) 535 N
E) 784 N



P08. A horizontal spring with a spring constant of 1800N/m is compressed by 28.0cm. A 5.60kg box is placed against the compressed spring. The system is released and the spring propels the box across the floor. The coefficient of friction between the box and the floor is 0.190. The box slides for 5.00m before impacting a wall. At what speed is the box moving when it hits the wall?
A) 2.56 m/s
B) 4.32 m/s
C) 5.02 m/s
D) 6.58 m/s
E) 8.44 m/s

P09. An 800.0kg car speeds around a curved road. The road is flat (not banked) and the coefficient of friction between the tires and the road is 0.820. If the radius of curvature of the road is 250.0m, then what is the maximum speed that the car can travel without sliding off the road?

A) 37.4 m/s
B) 44.8 m/s
C) 49.5 m/s
D) 64.9 m/s
E) 80.2 m/s



P10. A mass of 135.0g is attached to a spring that has a spring constant of 180.0N/m. The spring is stretched 20.0cm from equilibrium and released from rest. What is the speed of the mass when it passes back through the equilibrium point?

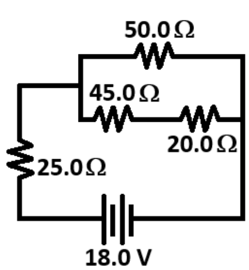
A) 1.33 m/s
B) 5.48 m/s
C) 7.30 m/s
D) 16.3 m/s
E) 26.7 m/s

P11. You found a meteorite and wish to determine its density. The meteorite weighs 11.2N in air, and when completely submerged in pure water, the meteorite weighs 9.40N. What is the density of the meteorite?

A) 1200 kg/m^3
B) 2500 kg/m^3
C) 3800 kg/m^3
D) 5200 kg/m^3
E) 6200 kg/m^3

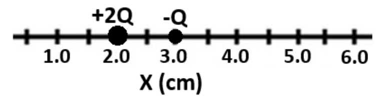
P12. For the DC circuit shown below, determine the power dissipated in the 45.0Ω resistor.

A) 0.760 W
B) 0.972 W
C) 2.03 W
D) 5.14 W
E) 6.61 W



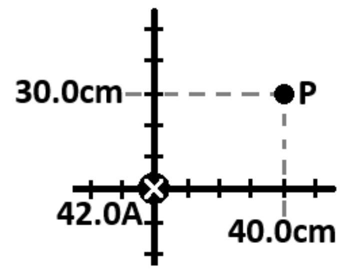
P13. Two point charges, $+2Q$ and $-Q$ are placed on a line as illustrated. The $+2Q$ charge is placed at the $x = 2.0$cm mark, and the $-Q$ charge is placed at the $x = 3.0$cm mark. At what point on the line is the electric field due to these charges exactly equal to zero?

A) $x = 1.0$ cm
B) $x = 2.7$ cm
C) $x = 3.5$ cm
D) $x = 4.0$ cm
E) $x = 5.4$ cm



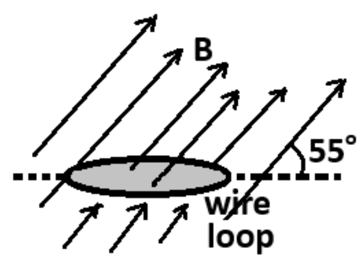
P14. A long, straight wire carries a current of 42.0A in the $-z$ direction. The wire lies along the z-axis and passes through the origin of our coordinate system, as shown. What is the horizontal component (the x-component) of the magnetic field at the point P (40.0cm, 30.0cm) due to the current carrying wire?

A) 10.1 μT
B) 13.4 μT
C) 16.8 μT
D) 21.0 μT
E) 28.0 μT



P15. A magnetic field of 5600μT is oriented at 55.0° with respect to the x-y plane, as shown. A circular loop of wire with a diameter of 40.0cm lies in the x-y plane. The loop has a resistance of 3.30Ω. The magnetic field decreases to zero in a time of 5.50ms. What is the magnitude of the current induced in the loop as the field changes?

A) 155 mA
B) 128 mA
C) 105 mA
D) 38.8 mA
E) 31.8 mA



P16. A 560.0nm laser is directed onto a double slit apparatus. The double slit apparatus is located 1.80m from a screen onto which an interference pattern is projected. The distance from the central maximum of the pattern to the tenth maximum is 1.65cm. What is the separation of the two slits in the double slit apparatus?

A) 10.7 μm
B) 33.9 μm
C) 61.1 μm
D) 339 μm
E) 611 μm

P17. Light of an unknown wavelength is directed onto a photoelectric metal. While illuminated, the metal ejects electrons from its surface with a velocity of 1.70×10^6 m/s. The workfunction of the metal is 1.93eV. What is the wavelength of the light shining on the metal surface?

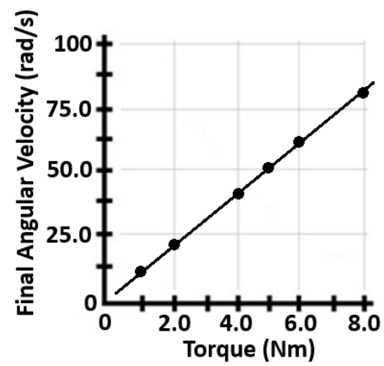
A) 67.5 nm
B) 75.5 nm
C) 122 nm
D) 151 nm
E) 197 nm

P18. The carbon-14 activity of a freshly cut 1.00-kilogram chunk of wood is 6.10nCi. An ancient totem made from 0.350kg of wood has a carbon-14 activity of 1.22nCi. How old is the wooden totem? The half-life of carbon-14 is 5730 years.

A) 3210 years
B) 4630 years
C) 5730 years
D) 9220 years
E) 13300 years

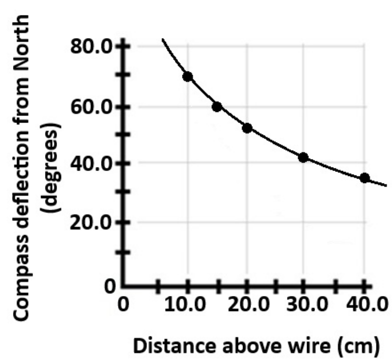
P19. You conduct an experiment in which a flywheel is accelerated by an adjustable torque. During each trial, the flywheel starts from rest and the torque acts on it for exactly 4.00 seconds. The final angular velocity of the flywheel is measured, and the data are plotted below. Based on these data, what is the rotational inertia of the flywheel?

A) 4.0 kgm^2
B) 2.0 kgm^2
C) 1.2 kgm^2
D) 0.40 kgm^2
E) 0.10 kgm^2



P20. A long straight wire carries a current of unknown magnitude directed due North. A compass is placed horizontally above the wire, and the deflection of the compass from true North is measured for different distances from the wire. The data are plotted below. Given that the horizontal component of the Earth's magnetic field in Texas is about 30μT, determine the approximate magnitude of the current flowing in the wire.

A) 10 A
B) 20 A
C) 30 A
D) 40 A
E) 50 A



Chemistry

1A 1	2A 2	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8	8B 9	8B 10	1B 11	2B 12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

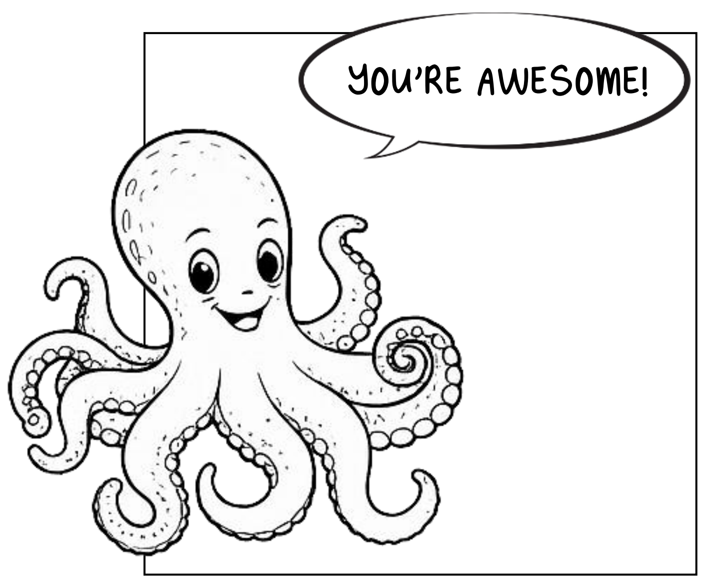
T_{mp} = 0°C
T_{bp} = 100°C
c_{ice} = 2.09 J/g·K
c_{water} = 4.184 J/g·K
c_{steam} = 2.03 J/g·K
ΔH_{fus} = 334 J/g
ΔH_{vap} = 2260 J/g
K_f = 1.86 °C/m
K_b = 0.512 °C/m

Constants

$R = 0.08206$ L·atm/mol·K
$R = 8.314$ J/mol·K
$R = 62.36$ L·torr/mol·K
$e = 1.602 \times 10^{-19}$ C
$N_A = 6.022 \times 10^{23}$ mol^{-1}
$k = 1.38 \times 10^{-23}$ J/K
$h = 6.626 \times 10^{-34}$ J·s
$c = 3.00 \times 10^{8}$ m/s
$R_H = 2.178 \times 10^{-18}$ J
$m_e = 9.11 \times 10^{-31}$ kg
$\mathscr{F} = 96{,}485$ C/mol e$^-$
1 amp = 1 C/sec
1 mol e$^-$ = 96,485 C

Liquid Densities

Density of ethanol at 25°C = 0.789 g/mL
Density of water at 25°C = 0.997 g/mL



Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	$9.80\ m/s^2$
Permittivity of Free Space	ε_0	$8.854 \times 10^{-12}\ C^2/Nm^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7}\ Tm/A$
Coulomb constant	k	$8.99 \times 10^{9}\ Nm^2/C^2$
Speed of light in a vacuum	c	$3.00 \times 10^{8}\ m/s$
Fundamental charge	e	$1.602 \times 10^{-19}\ C$
Planck’s constant	h	$6.626 \times 10^{-34}\ Js$
Electron mass	m_e	$9.11 \times 10^{-31}\ kg$
Proton mass	m_p	$1.67265 \times 10^{-27}\ kg$ $1.007276 amu$
Neutron mass	m_n	$1.67495 \times 10^{-27}\ kg$ $1.008665 amu$
Atomic Mass Unit	amu	$1.66 \times 10^{-27}\ kg$ $931.5\ MeV/c^2$
Gravitational constant	G	$6.67 \times 10^{-11}\ Nm^2/kg^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8}\ W/m^2K^4$
Universal gas constant	R	$8.314\ J/mol \cdot K$ $0.082057\ L \cdot atm/mol \cdot K$
Boltzmann’s constant	k_B	$1.38 \times 10^{-23}\ J/K$
Speed of Sound (at 20ºC)	v	343 m/s
Avogadro’s number	N_A	$6.022 \times 10^{23}\ atoms/mol$
Electron Volts	eV	$1.602 \times 10^{-19}\ J/eV$
Distance Conversion	miles → meters	1.00 mile = 1609 meters
Rydberg Constant	R_∞	$1.097 \times 10^{7}\ m^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^{5}\ Pa$
Density of Pure Water	ρ_{water}	$1000.0\ kg/m^3$

DO NOT DISTRIBUTE TO STUDENTS BEFORE OR DURING THE CONTEST!

UIL HIGH SCHOOL SCIENCE CONTEST
ANSWER KEY
2026 DISTRICT

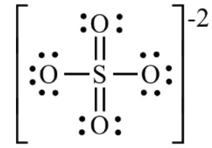
Biology		Chemistry		Physics	
B01.	C	C01.	C	P01.	D
B02.	A	C02.	D	P02.	C
B03.	E	C03.	C	P03.	B
B04.	D	C04.	B	P04.	C
B05.	A	C05.	A	P05.	A
B06.	B	C06.	C	P06.	B
B07.	B	C07.	B	P07.	A
B08.	E	C08.	B	P08.	A
B09.	B	C09.	E	P09.	B
B10.	D	C10.	D	P10.	C
B11.	B	C11.	E	P11.	E
B12.	C	C12.	B	P12.	B
B13.	A	C13.	D	P13.	E
B14.	A	C14.	E	P14.	A
B15.	C	C15.	A	P15.	E
B16.	E	C16.	D	P16.	E
B17.	B	C17.	A	P17.	C
B18.	D	C18.	D	P18.	B
B19.	D	C19.	E	P19.	D
B20.	A	C20.	C	P20.	D

CHEMISTRY SOLUTIONS – UIL DISTRICT 2026

C01. (C) Zinc forms 2+ ions and phosphate is PO_4^{3-}, so the neutral compound has three zinc ions and two phosphate ions, $Zn_3(PO_4)_2$, for a total of 13 atoms.

C02. (D) The reaction is $2AgNO_3(aq) + Na_2S(aq) \rightarrow 2NaNO_3(aq) + Ag_2S(s)$

C03. (C) $Z_{eff} = Z - S$ where Z equals the atomic number (the number of protons in the nucleus) and S equals the number of inner shell (non-valence) electrons.



C04. (B) The Lewis dot structure for a sulfate ion has two double bonds and two single bonds around the sulfur atom. The single bonds and the first bond in a multiple bond are σ bonds, and the second bond in a double bond is a π bond.

C05. (A) 50 × 0.997 = 49.85 g water. 50 × 0.789 = 39.45 g. total mass = 89.30 grams, total volume = 96.4 mL density = 89.30/96.4 = 0.92635 = 0.926 g/mL

C06. (C) PV = nRT so n = PV/RT = (1 atm)(85 L)/(0.08206)(296 K) = 3.4994 moles of gas. Since they filled half of the balloon with helium, half of the moles are helium (1.7497 moles He) and half are hydrogen (1.7497 moles H_2). The total mass of the gases is therefore 1.7497 × 4.00 + 1.7497 × 2.02 = 6.9988 + 3.5344 = 10.533 grams = 10.5 g.

C07. (B) 6.5 g / 84.01 g/mol = 0.07737 mol. 0.07737 mol × 28.5 kJ/mol = 2.205 kJ = 2200 J

C08. (B) This is a compound because the particles are made up of more than one different kind of atom. This is not a mixture because all of the particles (molecules in this case) are the same. This is a pure compound.

C09. (E) Melting ice is endothermic, so if the ice is melting it must be absorbing heat from its surroundings. The other statements are false because A) water is denser than ice, so the particles move closer together as the ice melts to water, B) temperature is a measure of the average kinetic energy of the particles, so if the temperature is the same the average KE (and therefore the speed of the molecules) is the same, C) the problem states that both are at 0°C, D) the molecules themselves do not change, it is only the trapped air and minerals in ice that make ice appear white.

C10. (D) First you need to know the concentration of each gas in moles per liter. Both of the reactants are 8 moles/1000 L = 0.008 M. The product is 16 moles/1000 L or 0.016 M. K_c for the reaction is

$$K_c = \frac{[\bigcirc]^2}{[\bullet][\circledcirc]} = (16)^2/(8)(8) = 256/64 = 4$$

(In this case the volume cancels out, so if a student tried to solve this problem just using moles instead of concentration they would still get the right answer, *but this will not always be the case*. For example, if the coefficients in the balanced equation were all 1's, using moles instead of concentration would give you the wrong answer.)

C11. (E) $K_a = [H^+][A^-]/[HA]$. From the pH we know that $[H+] = 10^{-pH} = 10^{-3.55} = 0.000282$ M. Since each H^+ created by the ionization of the acid also creates one A^-, $[A^-] = [H^+] = 0.000282$ M. [HA] = the initial concentration minus the amount dissolved = 0.50 M – 0.000282 M = 0.49972 M ≈ 0.50 M. $K_a = [0.000282][0.000282]/[0.50] = 1.59 \times 10^{-7}$.

C12. (B) $K_{sp} = [Ag^+]^3[PO_4^{3-}]$ $[PO_4^{3-}] = x$, $[Ag^+] = 3x$ $K_{sp} = (3x)^3(x) = 27x^4$
$x = (K_{sp}/27)^{1/4} = (8.9 \times 10^{-17}/27)^{1/4} = 4.26 \times 10^{-5}$ M. $[Ag^+] = 3x = 1.27 \times 10^{-4}$ M

C13. (D) The chemical formula is $H_4P_2O_6$. Each H atom is in a +1 oxidation state and each oxygen is in a –2 oxidation state. +4 – 12 = –8, so each phosphorus must be +4.

C14. (E) The reactants are A and B because their concentrations start high and drop as the reaction proceeds. The products are C and D because their concentrations increase as the reaction proceeds. The concentration of A drops twice as fast as the concentration of B, so the coefficient in front of A is twice the coefficient in front of B. Similarly, the concentration of D rises twice as fast as the concentration of C, so the coefficient in front of D is double the coefficient in front of C. The rate of decrease for A matches the rate of increase for D, so their coefficients are the same, and the same is true for B and C.

C15. (A) From the HNO_3: 0.070 L × 0.010 mol/L = 0.007 mol H^+. From the HCl: 0.100 L × 0.10 mol/L = 0.010 mol H^+. Total mol H+ = 0.010 + 0.007 = 0.017. Total volume = 0.100 + 0.070 = 0.170 L. $[H^+]$ = 0.06294 mol/L pH = –log (0.06294) = 1.20

C16. (D) Arrow A is the activation energy for the forward reaction. Arrow D is the activation energy for the reverse reaction. Since the D arrow is shorter than arrow A, the reverse reaction has a lower activation energy than the forward reaction and will therefore have a higher rate constant.

C17. (A) The equation for the reaction is $2Na(s) + Cl_2(g) \rightarrow 2\ NaCl(s)$ A reaction that has more than one reactant but only one product is a synthesis reaction.

C18. (D) The compound with the highest percent iron by mass is the one that has the lowest mass of anions. Since each of these compounds has one iron atom per formula unit, you just have to look at the masses of the anions and choose the one with the lowest mass. The compounds are $Fe(NO_3)_2$, $FeCl_2$, $Fe(NO_2)_2$, FeO, and $FeSO_4$. Since the oxide ion has a mass of 16 and the other are all well above that, iron(II) oxide has the highest percent of iron by mass.

C19. (E) Moles of bromate (BrO_3^-) = 1.050 L × 0.47 M = 0.4935 mol. Since $Ba(BrO_3)_2(s)$ requires 2 moles of bromate for every one mole of barium bromate, moles of $Ba(BrO_3)_2(s)$ = ½ × 0.4935 mol = 0.24675 mol. The molar mass of $Ba(BrO_3)_2$ = 393.13 g/mol × 0.24675 mol = 97.00 grams

C20. (C) 1 A = 1 C/sec, so 0.27 A × 500 sec = 135 C. 135 C / 96485 C/mol = 0.001399 moles of electrons. Since it takes one mole of electrons to reduce one mole of Ag^+ to Ag, 0.001399 moles of solid Ag will be formed. 0.001399 mol × 107.87 g/mol = 0.15093 g of Ag = 150 mg. (The significant digits in the current limit our answer to 2 significant digits.) The concentration and volume of the solution don't matter as long as there is enough Ag in the solution that it doesn't run out (1.0 L × 0.5 M = 0.5 mol Ag^+ which is way more than the 0.001399 moles we are depositing out).

PHYSICS SOLUTIONS – UIL DISTRICT 2026

P01. (D) page 88: "Over the next few years researchers in Canada, Austra, Germany, Sweden, and England observed 'Schweidler's fluctuations,' confirming that radioactivity was a random process that followed laws of probability."

P02. (C) pages 103-104: "In 1911, he proposed that the inseparable elements [mesothorium and radium] were not only similar, but identical. For nearly a century, the idea that each chemical element had a unique atomic weight had been chemical dogma. Soddy, who had already overthrown the belief in unchangeable elements with the transmutation theory, was now willing to discard the primacy of atomic weight."

P03. (B) page 110: "Thompson's assistant Francis W. Aston tried to separate the particles that created the different curves and measure their densities. One type was heavier than neon's accepted atomic weight, while the other was lighter. Yet their spectra were identical, and matched neon's spectrum. Apparently both types of particles were neon ions…"

P04. (C) The stars in this kind of binary system are close together – too close to visually separate. The only way the binary system can be identified is based on the Doppler shifting of the stellar spectral lines. The spectral lines shift red as the star's orbit moves it away from us, and the lines shift blue as the star's orbit moves it towards us. Since this motion is detected by observing the stellar spectrum, these binary star systems are called spectroscopic binaries.

P05. (A) Both *Newtons* and *Watts* are composite units, with a *Watt* being a *Joule per second*, and a *Joule* being a *Newton*meter*. Putting this together gives $x = \frac{Pz}{F} \;\; \rightarrow \;\; [m] = \frac{[W]z}{[N]} = \frac{[J]z}{[s][N]} = \frac{[N][m]z}{[s][N]} = \frac{[m]z}{[s]}$. This simplifies to $[m][s] = [m]z \;\; \rightarrow \;\; z = [s] = seconds$.

P06. (B) We first need to find the time that the football was in the air. To do this, we look at the vertical motion of the football by using $y_f = y_i + v_{iy}t + \frac{1}{2}a_y t^2$. The initial velocity component in the vertical direction is $v_{iy} = v_i \sin\theta = (17.0)\sin(22.0) = 6.368 m/s$. Now we put this together, giving $2.0m = 2.0m + (6.368 m/s)t + (0.5)(-9.80 m/s^2)t^2 \;\; \rightarrow \;\; 0 = 6.368t - 4.9t^2$. This simplifies to $4.9t = 6.368$, which gives a time of $t = 1.30s$. Now we find the horizontal distance down the field where the ball was caught by using: $x_f = x_i + v_{ix}t + \frac{1}{2}a_x t^2$. There is no horizontal acceleration, and the horizontal component of the initial velocity is $v_{ix} = v_i\cos\theta$. Thus, we get $x_f = 0 + v_i\cos\theta\, t + 0 \;\; \rightarrow \;\; x_f = (17.0 m/s)\cos(22.0)\,(1.30s) = 20.5m$.

P07. (A) Let's begin by labeling the tensions: T_1 is the cable on the left and is already marked for us. Let's call the tension in the cable on the upper right T_2, and the tension in the vertical rope T_3. The system is in equilibrium, so all forces will sum to zero. Looking at the dangling mass, we have two forces, and both are vertical: mg is directed downward and T_3 is directed upward. Summing these gives
$\sum F_m = T_3 - mg = 0 \;\; \rightarrow \;\; T_3 = mg = (40.0kg)(9.80 m/s^2) = 392N$.
Now we focus our attention on the connection point between the rope and the cables. The tensions in the cables are at angles, so they must be broken into components. For T_2 we get components $T_{2x} = T_2\cos(45.0°)$ (right, positive) and $T_{2y} = T_2\sin(45.0°)$ (up, positive). For T_1 we get components $T_{1x} = -T_1\cos(30.0°)$ (left, negative) and $T_{1y} = T_1\sin(30.0°)$ (up, positive). With respect to the connection point, the tension T_3 points down, so it is negative and entirely in the y-direction: $T_3 = T_{3y} = -392N$. Now consider the sums of forces. For the x-direction, the sum of the forces acting on the connection point are: $\sum F_x = T_2\cos(45.0°) - T_1\cos(30.0°) = 0$. This leads to $T_2(0.7071) = T_1(0.8660) \;\; \rightarrow \;\; T_2 = 1.225T_1$. Now, summing the forces in the y-direction gives $\sum F_y = T_2\sin(45.0°) + T_1\sin(30.0°) - 392 = 0$. Using the substitution found from the x-direction, we get $1.225T_1\sin(45.0°) + T_1\sin(30.0°) - 392 = 0 \;\; \rightarrow \;\; 0.8660T_1 + 0.5T_1 = 392$.
This gives $1.366T_1 = 392 \;\; \rightarrow \;\; T_1 = 287N$.

P08. (A) This problem is best solved using conservation of energy along with the work done by the frictional force. To calculate work, we do need to consider the forces and the free body diagram for the box. While it is freely sliding, there are three forces acting on the box: gravity (mg, directed downward), the

normal force (F_N, directed upward), and friction (F_f, directed horizontally opposite to the motion). The box does not accelerate in the vertical direction, so the vertical forces must sum to zero: $\sum F_y = F_N - mg = 0$. This gives $F_N = mg = (5.60)(9.80) = 54.88N$. From this we obtain the frictional force: $F_f = \mu F_N = (0.190)(54.88) = 10.43N$. Now that we have the frictional force, we can consider the energy of the system. Initially, the spring is compressed, and the box is stationary – the initial energy is entirely in the form of elastic potential energy. When the box reaches the wall, the spring is uncompressed, the box is moving, and friction has converted some energy to heat. Thus, the final energy consists of the kinetic energy of the box and the work done by friction. Mathematically, we have $E_i = EPE_i = \frac{1}{2}kx_i^2$ and $E_f = KE_f + W_f = \frac{1}{2}mv_f^2 + F_f d$. By conservation of energy, these are equal: $E_i = E_f \;\; \rightarrow EPE_i = KE_f + W_f$. This leads to $\frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + F_f d$. Plugging in the known values, we find the velocity of the box when it reaches the wall:
$\frac{1}{2}\left(\frac{1800N}{m}\right)(0.280m)^2 = \frac{1}{2}(5.60kg)v_f^2 + (10.43N)(5.00m) \;\rightarrow\; 70.56 = 2.80v_f^2 + 52.15 \;\rightarrow$ $18.41 = 2.80v_f^2$. This gives a final velocity of $v_f = 2.56m/s$.

P09. (B) First, we consider the free body diagram for the car. There are three forces acting on the car: gravity (mg, directed downward), the normal force (F_N, directed upward), and friction (F_f, directed horizontally towards the center of the circular curve of the road). The car has no motion in the vertical direction, so the vertical forces must sum to zero. This gives $\sum F_y = F_N - mg = 0$. Thus, $F_N = mg$. From this we can obtain the frictional force: $F_f = \mu F_N = \mu mg$. Now, the sum of all forces directed towards the center of the circular motion must equal the centripetal force. Only the frictional force points in this direction, so we equate the frictional force to the centripetal force: $F_c = \sum F_{to\ center} = F_f \;\rightarrow\; F_c = \frac{mv^2}{r} = F_f = \mu mg$. This gives $\frac{mv^2}{r} = \mu mg \;\;\rightarrow\;\; v^2 = \mu r g = (0.820)(250.0m)(9.80m/s^2) = 2009$. So, the maximum speed that the car can travel without slipping is $v = \sqrt{2009} = 44.8\ m/s$.

P10. (C) Although a mass-spring system is an oscillatory system, we don't require any oscillation information to solve this problem - we simply use conservation of energy. Initially, the energy is entirely in the form of the elastic potential energy of the stretched spring. When the mass passes through the equilibrium point, the energy has been converted completely into the kinetic energy of the mass. By conservation of energy, we get: $\frac{1}{2}kA^2 = \frac{1}{2}mv^2$. Solving for the velocity, we obtain $kA^2 = mv^2 \;\;\rightarrow\;\; v^2 = \frac{k}{m}A^2 \;\;\rightarrow$
$v = A\sqrt{\frac{k}{m}} = (0.200m)\sqrt{\frac{180.0N/m}{0.135kg}} = 7.30m/s$.

P11. (E) The weight of the meteorite in air can be expressed in terms of the density and volume of the meteorite: $W_{air} = M_m g = \rho_m V_m g = 11.2N$. When placed in water, there is also an upward buoyant force acting on the meteorite, leading to a smaller apparent weight. The difference between the weight in water and the weight in air equals the magnitude of the buoyant force:
$F_B = 11.2N - 9.40N = 1.8N$. The buoyant force is given by $F_B = \rho_{liquid} V_m g$. This leads to $F_B = \rho_{liquid} V_m g = 1.8N \;\;\rightarrow\;\; (1000.0kg/m^3)V_m(9.80) = 1.8 \;\;\rightarrow\;\; V_m = 1.837 \times 10^{-4}m^3$. This is the volume of the meteorite. Plugging this into our "weight in air" formula gives:
$\rho_m V_m g = 11.2N = \rho_m(1.837 \times 10^{-4}m^3)(9.80) = 11.2N$. Thus, the density of the meteorite is calculated to be $\rho_m = 6200kg/m^3$.

P12. (B) To begin, let's label the resistors. Let $R_1 = 25.0\Omega$, $R_2 = 50.0\Omega$, $R_3 = 45.0\Omega$, and $R_4 = 20.0\Omega$. Now, we combine the resistors into a single equivalent resistance. The first two that we combine are the 45.0Ω and 20.0Ω in series: $R_A = R_3 + R_4 = 45.0 + 20.0 = 65.0\Omega$. This resistance is then combined in parallel with the 50.0Ω resistor: $R_P = (R_A^{-1} + R_2^{-1})^{-1} = ((65.0)^{-1} + (50.0)^{-1})^{-1} = 28.26\Omega$. Finally, we combine this group in series with the 25.0Ω resistor: $R_T = R_P + R_1 = 28.26 + 25.0 = 53.26\Omega$. This is the total equivalent resistance of the circuit. Now we find the current flowing from the battery: $V_T = I_T R_T \;\;\rightarrow$ $18.0V = I(53.26\Omega) \;\;\rightarrow\;\; I_T = 0.3380A$. Since they are in series, this same current flows through both the 25.0Ω resistor and the parallel group equivalent resistance: $I_T = I_1 = I_P = 0.3380A$. Next, we find the voltage across the parallel group: $V_P = I_P R_P = (0.3380)(28.26) = 9.55V$. By parallel rules, this voltage must be the same across the top and bottom branches of the parallel group.

Mathematically: $V_P = V_2 = V_A = 9.55V$. From this we find the current flowing in the lower branch of the parallel group: $V_A = I_A R_A \;\rightarrow\; 9.55V = I_A(65.0\Omega) \;\rightarrow\; I_A = 0.1469A$. Again, since they are in series, this same current flows in both the 45.0Ω and 20.0Ω resistors: $I_A = I_3 = I_4 = 0.1469A$. Finally, we calculate the energy dissipated in the 45.0Ω resistor: $P_3 = I_3^2 R_3 = (0.1469)^2(45.0) = 0.972W$.

P13. (E) Electric fields are vectors, so we are looking for a place where the electric field magnitude due to each charge is the same value, but the directions of the electric fields from each charge are opposite one another. This can only happen to the left of the 2Q charge or to the right of the −Q charge. Between the two charges, their fields are oriented in the same direction. The magnitudes of the electric fields produced by the two charges cannot be the same to the left of the 2Q charge, because the left region is closer to the larger magnitude charge. The field produced by the larger charge will never be matched by the more distant small charge. Thus, we have determined that the point where the electric field is zero will be to the right of the −Q charge. Let that point be a distance y from the −Q charge; and, consequently, a distance of $1 + y$ from the 2Q charge. The magnitude of the electric field from the 2Q charge is $|E_1| = \left|\frac{k(2Q)}{(1+y)^2}\right|$, and the magnitude of the electric field from the −Q charge is $|E_2| = \left|\frac{k(-Q)}{(y)^2}\right|$. We want these fields to be of equal magnitude: $\left|\frac{k(2Q)}{(1+y)^2}\right| = \left|\frac{k(-Q)}{(y)^2}\right|$. This simplifies to $\frac{2}{(1+y)^2} = \frac{1}{y^2} \;\rightarrow\; 2y^2 = 1 + 2y + y^2 \;\rightarrow\; y^2 - 2y - 1 = 0$. This quadratic has solutions $y = \frac{2\pm\sqrt{8}}{2} = 1 \pm \sqrt{2}$. The solution $y = 1 - \sqrt{2}$ is negative, so it makes no sense since y is a distance. Thus, the solution is $y = 1 + \sqrt{2} = 2.414$. This is a distance to the right of the −Q charge, which means that the location on the line at which the electric field is zero is $x = 3.0 + y = 3.0 + 2.414 = 5.414 \approx 5.4\text{cm}$.

P14. (A) First, we find the magnitude of the magnetic field produced at the point P by the current-carrying wire. The distance from the wire (at the origin) to the point P is $r = \sqrt{(40.0cm)^2 + (30.0cm)^2} = 50.0cm = 0.500m$. Thus, the magnitude of the magnetic field at the point P is $|B| = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7})(42.0)}{2\pi(0.500)} = 1.68 \times 10^{-5}T = 16.8\mu T$. The magnetic field is oriented in circles around the long, straight wire (the field direction is determined by a vector cross product). For a $-z$ current direction, the right-hand rule gives the direction of the magnetic field to be clockwise in circles around the wire. Another result of the cross product is that the trigonometric functions are flipped from what we might expect. Using a little geometry, we find that the horizontal component of the magnetic field is $B_x = |B|\sin\theta$. Using the triangle to calculate the sine, we obtain $B_x = |B|\frac{opposite}{hypotenuse} = (16.8\mu T)\frac{30.0cm}{50.0cm} = 10.1\mu T$.

P15. (E) First, we find the magnetic flux passing through the loop. The flux is given by $\Phi_B = BA\cos\theta$ where A is the area of the wire loop and θ is the angle between the magnetic field direction and the normal line to the face of the loop. From what we are given, we can easily find both:
$A = \pi r^2 = \pi\left(\frac{d}{2}\right)^2 = \pi\left(\frac{0.400}{2}\right)^2 = \pi(0.200m)^2 = 0.1257m^2$, and
$\theta = 90 - \phi = 90.0 - 55.0 = 35.0°$. Now we find the magnetic flux:
$\Phi_B = BA\cos\theta = (5600 \times 10^{-6}T)(01257m^2)\cos(35.0°) = 5.76 \times 10^{-4}Tm^2$. This is the initial flux (Φ_{Bi}); after the field vanishes, the final flux is zero ($\Phi_{Bf} = 0.0$). The *EMF* (voltage) produced by the changing magnetic flux is given by Faraday's Law:
$\mathcal{E} = -\frac{\Delta\Phi_B}{\Delta t} = -\frac{(\Phi_{Bf}-\Phi_{Bi})}{\Delta t} = -\frac{(0.0-5.76 \times 10^{-4}Tm^2)}{5.50 \times 10^{-3}s} = 0.1048V$. Lastly, we get the induced current by using Ohm's law: $\mathcal{E} = IR \;\rightarrow\; 0.1048V = I(3.30\Omega) \;\rightarrow\; I = 0.03176A = 31.8mA$.

P16. (E) The equation governing double-slit interference is $d\sin\theta = m\lambda$, where d is the separation of the slits and λ is the wavelength of the laser. To find the angle, we use the geometry of the interference pattern: $\tan\theta = \frac{X}{L} = \frac{1.65 \times 10^{-2}m}{1.80m} = 0.009167$. This gives $\theta = 0.525° \;\rightarrow\; \sin\theta = 0.009166$. Notice how $\sin\theta$ is almost exactly equal to $\tan\theta$ - this happens for very small angles, and the approximation $\sin\theta = \tan\theta$ is called the 'small-angle' approximation. Moving on, we notice that the distance X was for ten fringes, so we use $m = 10$ in the interference equation. Putting it all together, we obtain $d\sin\theta = m\lambda \;\rightarrow\; d(0.009166) = (10)(560.0nm) \;\rightarrow\; d = 610900nm = 610.9\mu m \approx 611\mu m$.

P17. (C) The electrons are moving at a non-relativistic speed, so we can use classical equations. Let's begin with the kinetic energy of the ejected electrons, given by:

$E_e = \frac{1}{2}mv^2 = (0.5)(9.11 \times 10^{-31}kg)\left(1.70 \times 10^6 \frac{m}{s}\right)^2 = 1.316 \times 10^{-18}J$. Converting this kinetic energy to eV, we get $E_e = \frac{1.316 \times 10^{-18}J}{1.602 \times 10^{-19}} = 8.217eV$. Now we find the energy of the photons illuminating the metal surface by using the photoelectric equation: $E_e = E_\gamma - \phi$ where ϕ is the workfunction of the metal. This gives: $8.217eV = E_\gamma - 1.93eV \;\; \rightarrow \;\; E_\gamma = 10.15eV$. To find the wavelength of these photons, we use the handy relation: $\lambda = \frac{1240eVnm}{E_\gamma} = \frac{1240eVnm}{10.15eV} = 122nm$.

P18. (B) Since 1.00kg of freshly cut wood has an activity of 6.10nCi, we would expect the activity of 0.350kg of freshly cut wood to have an activity of $A_0 = (0.350kg)\frac{6.10nCi}{1.00kg} = 2.135nCi$. Now, to find the age of the wooden totem, we first need the decay constant of carbon-14:
$\lambda = \frac{\ell n(2)}{T_{1/2}} = \frac{0.693}{5730years} = 1.209 \times 10^{-4}yr^{-1}$. We then use the decay formula: $A = A_0 e^{-\lambda t} \;\; \rightarrow$
$1.22nCi = 2.135nCie^{-(1.209 \times 10^{-4}yr^{-1})t}$. This leads to $\ell n\left(\frac{1.22}{2.135}\right) = -(1.209 \times 10^{-4}yr^{-1})t \;\; \rightarrow$
$0.5596 = (1.209 \times 10^{-4})t \;\; \rightarrow \;\; t = 4627\, years \; \approx 4630\, years$. This is the age of the totem.

P19. (D) The relation between torque and motion begins with angular acceleration, and the equation $\tau = I\alpha$, where I is the rotational inertia. Angular acceleration relates to the final angular velocity by the kinematic equation: $\omega_f = \omega_i + \alpha t$. Since the flywheel starts from rest for each trial, and since the torque acts for exactly 4.00seconds, we can simplify this equation:
$\omega_f = \omega_i + \alpha t = 0 + \alpha(4) \;\; \rightarrow \;\; \omega_f = 4\alpha$, or $\alpha = \frac{1}{4}\omega_f$. Combining this with the torque equation gives: $\tau = I\alpha = I(\frac{1}{4}\omega_f)$. Solving for the final angular velocity, we obtain $\omega_f = \frac{4}{I}\tau$. This expression is linear, and the slope of the line equals $\frac{4}{I}$. Turning to our data, we see that it is linear. To find the slope, I will choose two points that are on the line but are not data points: (2.5Nm, 25.0rad/s) and (7.2Nm, 75.0rad/s). These give a slope of $(slope) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{75.0-25.0}{7.2-2.5} = 10.6$. Equating this to the expression derived earlier leads to: $(slope) = 10.6 = \frac{4}{I} \;\; \rightarrow \;\; I = 0.376 \approx 0.40kgm^2$.

P20. (D) Because the wire carries a Northward current, the magnetic field generated above the wire will be entirely in the Eastward direction (determined by using the right-hand rule). Notably, the magnetic field produced by the current directly above the wire is perpendicular to the horizontal component of the Earth's magnetic field. This allows us to relate the deflection of the compass to the strength of the produced field: $\tan\theta = \frac{B_{current}}{B_{Earth-horiz}}$. Plugging in the magnitude of the horizontal component of the Earth's magnetic field, we obtain $\tan\theta = \frac{B_{current}}{30\mu T} \;\; \rightarrow \;\; B_{current} = (30\mu T)\tan\theta$. The magnetic field produced by a long, straight, current-carrying wire is given by:

$B_{current} = \frac{\mu_0 I}{2\pi R} = \frac{(4\pi \times 10^{-7}Tm/A)I}{2\pi R} = (2.0 \times 10^{-7}Tm/A)\frac{I}{R}$. By converting this into μT, we obtain: $B_{current} = (2.0 \times 10^{-1}\mu Tm/A)\frac{I}{R}$. Equating these two expressions for the magnetic field produced by the current gives: $B_{current} = (30\mu T)\tan\theta = (0.2\mu Tm/A)\frac{I}{R} \;\; \rightarrow \;\; (150A/m)R\tan\theta = I$. The plot is not linear, so we simply pick a data point to use to estimate the answer. I'll use (10.0cm, 70.0°). Using this data point, we obtain a current of $I = (150A/m)(0.10m)\tan(70°) = 41A \approx 40A$. Note: The other data points give comparable results.